

EDUCATION

The Ohio State University

Ohio, USA

Doctoral Student in Department of Computer Science and Engineering (on leave), GPA: 3.90/4.00

Aug. 2021–Dec. 2023

- **Advisor:** Dr. Wei-Lun Chao
- **Awards:** University Fellowship in 2021
- Completed 29 credit hours of courses, including Artificial Intelligence, Data Mining, and Real-time Rendering.
- Published 4 research papers at top AI/ML/CV conferences (ICLR, AACL, CVPR, NeurIPS).

National Taiwan University

Taipei, Taiwan

B.Sc. in Department of Computer Science and Information Engineering, GPA: 3.97/4.30

Sep. 2015–Jan. 2018

- **Advisor:** Dr. Jane Yung-jen Hsu

National Taiwan University

Taipei, Taiwan

B.Sc. in Department of Computer Science and Information Engineering, GPA: 4.02/4.30

Sep. 2011–Jun. 2015

- **Awards:** The Presidential Award (5%) in Fall 2012 and Fall 2013

EXPERIENCE

Senior Machine Learning Scientist

Taipei, Taiwan

Appier (沛星互動科技股份有限公司)

Mar. 2024–Jun. 2025

- Maintained and enhanced an LLM-powered [Knowledge Bot](#) with Retrieval-Augmented Generation (RAG) that integrated business domain knowledge to enable intelligent and context-aware customer service.

Graduate Research Assistant

Ohio, USA

Department of Computer Science and Engineering, The Ohio State University

Aug. 2021–Dec. 2023

- Holistic Transfer — A Practical Adaptation Setting with Partial Target Data ([NeurIPS23](#), [NeurIPS24](#))
- Visual Query Tuning — Parameter and Memory Efficient Fine-tuning for Vision Transformers ([CVPR23](#))
- Synthetic (Fractal) Data Generation and Pre-training ([AAAI23](#), [ICLR23](#))

Research Assistant

Taipei, Taiwan

AINTU Center, National Taiwan University

Jan. 2021–Aug. 2021

- Binary Hash Code Learning for Efficient Image Retrieval ([TAI21](#), [TNNLS22](#))
- Continual Learning for Defect Detection ([INDIN21](#))

Research Assistant

Taipei, Taiwan

Institute of Information Science, Academia Sinica

Apr. 2019–Dec. 2020

- Continual Learning ([NeurIPS19](#)), Network Compression ([IJCNN20](#))

PUBLICATIONS

* indicates equal contributions

Conferences

- [C12] Z. Mai, P. Zhang, **C.-H. Tu**, H.-Y. Chen, Q.-H. Nguyen, L. Zhang, and W.-L. Chao, "Lessons learned from a unifying empirical study of parameter-efficient fine-tuning (peft) in visual recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [C11] Z. Mai*, A. Chowdhury*, P. Zhang*, **C.-H. Tu**, H.-Y. Chen, V. Pahuja, T. Berger-Wolf, S. Gao, C. Stewart, Y. Su, and W.-L. Chao, "Fine-tuning is fine, if calibrated," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024, pp. 136 084–136 119.

- [C10] H.-Y. Chen, **C.-H. Tu**, Z. Li, H.-W. Shen, and W.-L. Chao, "On the importance and applicability of pre-training for federated learning," in *International Conference on Learning Representations (ICLR)*, 2023.
- [C9] **C.-H. Tu**^{*}, H.-Y. Chen^{*}, D. Carlyn, and W.-L. Chao, "Learning fractals by gradient descent," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2023.
- [C8] **C.-H. Tu**^{*}, H.-Y. Chen^{*}, Z. Mai, J. Zhong, V. Pahuja, T. Berger-Wolf, S. Gao, C. Stewart, Y. Su, and W.-L. Chao, "Holistic transfer: Towards non-disruptive fine-tuning with partial target data," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023, pp. 29 149–29 173.
- [C7] **C.-H. Tu**^{*}, Z. Mai^{*}, and W.-L. Chao, "Visual query tuning: Towards effective usage of intermediate representations for parameter and memory efficient transfer learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 7725–7735.
- [C6] C.-H. Chen, **C.-H. Tu**, J.-D. Li, and C.-S. Chen, "Defect detection using deep lifelong learning," in *2021 IEEE 19th International Conference on Industrial Informatics (INDIN)*, 2021, pp. 1–6.
- [C5] **C.-H. Tu**, J.-H. Lee, Y.-M. Chan, and C.-S. Chen, "Pruning depthwise separable convolutions for mobilenet compression," in *IEEE International Joint Conference on Neural Networks (IJCNN)*, 2020, pp. 1–8.
- [C4] **C.-H. Tu**, C.-E. Wu, and C.-S. Chen, "Extending conditional convolution structures for enhancing multitasking continual learning," in *Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, 2020, pp. 1605–1610.
- [C3] S. C.-Y. Hung, **C.-H. Tu**, C.-E. Wu, C.-H. Chen, Y.-M. Chan, and C.-S. Chen, "Compacting, picking and growing for unforgetting continual learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019, pp. 13 669–13 679.
- [C2] **C.-H. Tu**, C.-Y. Yang, and J. Y.-j. Hsu, "Idennet: Identity-aware facial action unit detection," in *IEEE International Conference on Automatic Face & Gesture Recognition (FG)*, 2019, pp. 1–8.
- [C1] H.-F. Yang, **C.-H. Tu**, and C.-S. Chen, "Adaptive labeling for hash code learning via neural networks," in *IEEE International Conference on Image Processing (ICIP)*, 2019, pp. 2244–2248.

Journals

- [J3] H.-F. Yang, **C.-H. Tu**, and C.-S. Chen, "Learning binary hash codes based on adaptable label representations," *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, vol. 33, no. 11, pp. 6961–6975, 2022.
- [J2] **C.-H. Tu**, H.-F. Yang, S.-M. Yang, M.-C. Yeh, and C.-S. Chen, "Semantichash: Hash coding via semantics-guided label prototype learning," *IEEE Transactions on Artificial Intelligence (TAI)*, vol. 2, no. 1, pp. 42–57, 2021.
- [J1] H.-F. Yang, T.-Y. Chen, **C.-H. Tu**, and C.-S. Chen, "Equivalent scanning network of unpadded cnns," *IEEE Signal Processing Letters*, vol. 25, no. 10, pp. 1590–1594, 2018.

COMPETITIONS

- **3rd Place, Efficient Online Continual Visual Instruction Tuning (CVIT) Challenge** Oct. 2025
ICCV 2025 6th CLVision Workshop
 Implemented LoRA-based parameter-efficient adaptation for LLaVA vision-language models. Developed and evaluated replay sampling strategies to mitigate forgetting in generative online continual learning with streaming data.

INVITED TALKS

- **Compacting, Picking and Growing for Unforgetting Continual Learning** Dec. 2020
AI Forum 2020 at Howard Civil Service International House, Taipei, Taiwan

PROFESSIONAL ACTIVITIES

- **Conference Reviewer:** ICML 2023-2025, NeurIPS 2023-2025, AAAI 2026, ICLR 2026
- **Journal Reviewer:** Pattern Recognition 2020

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, L^AT_EX
- **Development Tools:** UNIX, PyTorch, Tensorflow, Scikit-learn, Git